

Spatial expressions with implicit (and overt) arguments: A comparative study of French and Spanish

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Abstract: Implicit arguments in the prepositional domain are a lesser-studied context of NULL or NOTHING, yet much like in the inflectional and verbal domains, micro-variation within Romance reveals different types of implicit arguments and illuminates properties of the structures within which they are licensed. This chapter focuses on the null pronouns of orphan prepositions in French and Spanish and scrutinizes their licensing through the lens of a three-way typology of spatial expressions. It is shown that French and Spanish converge in more ways than they differ in terms of orphaning, and that the typology of spatial expressions is highly predictive for the licensing of null pronouns.

Key words: spatial prepositions; orphan prepositions; empty categories; implicit arguments; null objects; French; Spanish

1 Introduction

Taking French and Spanish as points of comparison, this study examines different types of implicit arguments in the prepositional domain, a lesser-studied context of NULL or NOTHING.¹ We demonstrate that the distribution of orphaning across spatial expressions is more similar than previously assumed and that both categoric and variable behaviour is predictable within a three-way typology of spatial expressions (SPEX).

The term SPEX is used in this chapter to refer to any expression which locates a Figure in relation to a Ground, including simple prepositions such as *on* (e.g., [*the cat*]_{FIGURE} *on* [*the mat*]_{GROUND}) and more complex forms such as *in front of* (e.g., [*the cat*]_{FIGURE} *in front of* [*the mat*]_{GROUND}). From a comparative perspective SPEX are interesting because, for example, morphologically complex SPEX in Spanish (e.g., *encima* 'on top') and French (e.g., *en face* 'in front') appear to pattern similarly with respect to implicit arguments while simple prepositions (e.g., *con/avec* 'with') are much more restrictive in Spanish. This kind of variation within Romance affords an opportunity to better understand the types of implicit arguments available in French and Spanish and presents an ideal testing ground for the predictive power of a typology of prepositional elements with respect to the licensing of implicit arguments.

Our approach is primarily descriptive: we bring the Spanish facts to bear on what is known about French. Our descriptive tools are theoretically informed by a typology of empty categories, foundational to the generative tradition, and by a cartographic view of the extended prepositional phrase as proposed in Svenonius (2010 et seq.). This chapter takes as its focus prepositions that license null pronominal elements, otherwise known as *orphan prepositions* in the French tradition (Zribi-Hertz 1984a,b), as this is the most prominent locus of variation within French and across languages. Zribi-Hertz's seminal work on orphan prepositions has inspired 40 years of scholarship, yet, to our knowledge, there has been no concerted micro-

¹ In this chapter we use "implicit argument" as a pre-theoretical cover term describing the absence of an overt argument. Where needed we will use more specific terms such as null pronoun, *pro*, ellipsis etc. (see also Section 2).

comparative work within Romance, an area rich with empirical and theoretical potential as we hope to show.

Section 2 introduces a distinction between three types of implicit arguments in the prepositional domain: unprojected implicit arguments; and two types of projected implicit arguments, namely, null pronouns and ellipsis. Section 3 then reviews the basic facts regarding orphan prepositions in French and Spanish. We also present in this section our analytic lens: a three-way typology of SPEX. In Section 4, we identify key points of convergence and divergence between these two closely related languages as regards the licensing of a null pronoun (*pro*) with SPEX.

2 Types of implicit arguments in PPs

In this section, we distinguish between three types of implicit arguments in PPs: unprojected implicit arguments, null pronouns (*pro*), and ellipsis.² While the former does not present evidence of a syntactically projected argument, we consider null pronouns and ellipsis to be two subtypes of syntactically projected implicit arguments.

The distinction between projected and unprojected implicit arguments in French PPs is proposed in Authier (2016) using common diagnostics for the pronominal status of implicit arguments: the availability of sloppy readings and the availability of referential dependency on a quantificational element. Authier takes implicit arguments which do not pass these tests as unprojected; i.e., cases of syntactic intransitivity. Among the implicit arguments that are projected, we further distinguish between null pronouns and ellipsis. For example, the implicit arguments of contrastively focused prepositions are best analysed as cases of ellipsis. This yields the taxonomy in Figure 1, which will be further motivated with French data in the remainder of this section. In line with Zribi-Hertz's (1984a,b) original proposal, we limit the term *orphan preposition* to cases where the implicit argument is a null pronoun.³

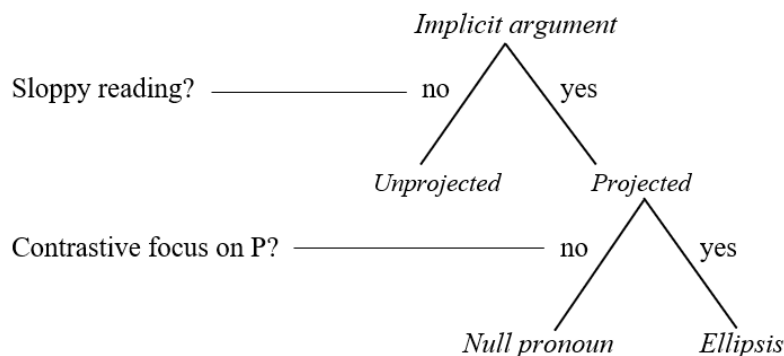


Figure 1: A taxonomy of implicit arguments in PPs (based on French and Spanish)

Starting with the first distinction of this taxonomy, Authier argues that prepositions are not a uniform class with respect to the status of the implicit argument. For *pour* 'for', Authier proposes that in some cases the complement position does not appear to be syntactically projected. His main argument involves the unavailability of sloppy readings; in (1a), the implicit argument of *pour* has a strict interpretation; it cannot be interpreted as referring to Sophie's aunt. This distinguishes *pour* from *avec* 'with', for which the implicit arguments can

² Traces (or copies) are another type of projected implicit argument, but they are less relevant in the prepositional domain for standard varieties of French and Spanish for which PPs are syntactic islands. See Roberge & Rosen (1999), Poplack et al. (2012), Roberge (2012) and references therein for discussion of stranding in North American varieties of French.

³ Both French and Spanish allow generic, definite, and deictic interpretations of *pro*, discussed below.

be ambiguous between a strict and a sloppy reading (see (1b)).⁴ The implicit argument of *avec* behaves in this respect like a clitic pronoun, which also allows both strict and sloppy readings (see Authier 2016: 241).

- (1) a. Quand Sylvie_i ne vote pas pour sa_i tante, Sophie ne vote pas pour Ø non plus.
'When Sylvie doesn't vote for her aunt, Sophie doesn't vote for her either.'
(Ø ≠ Sophie's aunt)
(Authier 2016: 261)
- b. Mon mari_i ne touche jamais à ses_i outils alors que le tien_j bricole avec
Ø toute la journée.
'My husband never touches his tools while yours tinkers with them all day
long.'
(Ø = his_i or his_j tools)
(Authier 2016: 243)

Authier (2016) takes these empirical observations not only to indicate that the implicit argument of prepositions like *avec* is projected, but that they are null pronouns. However, sloppy identity readings are not exclusive to null pronouns; they also occur in case of ellipsis.⁵ In fact, Gabriel (2002: 41) suggests a deletion analysis of implicit arguments (the complement is fully projected, but not phonologically realized, see (2)), and Kaiser (2012: 241) describes deletion as the "common analysis" for implicit arguments of prepositions in French.

- (2) La voiture n'est pas [PP devant [DP la maison]_i], elle est [PP derrière [DP ~~la maison~~]_i]
'The car is not in front of the house, it is behind it.'
(Gabriel 2002: 41)

Heidinger (2024) points out, however, that an ellipsis analysis for all implicit arguments is too strong. In French, the implicit argument of one and the same preposition can have different statuses depending on the syntactic-pragmatic context. For *avec* and *sans*, he argues that in full sentences without narrow contrastive focus on the preposition (as in (1b)) the implicit argument is a null pronoun. In fragments with a contrastive focus on the preposition (as in (3)), the implicit argument is elliptical, i.e., the result of deletion.⁶

- (3) Ils revinrent tous les week-ends. Avec ou sans Philibert_i. Plutôt avec Ø_i.
'They came back every weekend. With or without Philibert. Rather with him.'
(Gavalda, 2004, *Ensemble, c'est tout*; Frantext, modified)

Heidinger's (2024) empirical argument for this assumption is that in fragments such as (3), the implicit argument does not underly the same animacy restrictions as in non-contrastive full sentences. Starting with Zribi-Hertz (1984a,b) it has been frequently assumed that null pronominal complements of orphan prepositions tend to be used for (and are interpreted as)

⁴ The prospect of intransitive prepositions as Authier describes them merits more investigation, since, in simpler contexts without quantification, the implicit argument of *pour* easily allows sloppy readings.

(i) Max_i ne vote jamais pour son_i équipe alors que Marie_j vote toujours pour Ø.
'Max never votes for his team while Marie always votes for {it/hers}.'

⁵ As pointed out by a reviewer there is a long-standing debate concerning sloppy identity readings and its interpretation with respect to the status of implicit elements. We cannot do this literature justice here but refer the reader to Hoji (1998, 2003), Kim (1999), Landau (2010, 2018), and Martínez García (2023) and the discussion therein. As for our own proposal, we follow Authier (2016) who highlights the parallelism between French implicit arguments allowing sloppy readings and clitic (object) pronouns in the verbal domain which generally allow for sloppy readings as well.

⁶ C-selection is unaffected as both types of implicit arguments are DPs.

non-human referents, but both experimental and corpus studies have confirmed that the link between complement type and humanness is at best a strong tendency (see Heidinger 2023, 2024 for recent overviews, also Zribi-Hertz 2000). While the humanness of the complement has a strong effect on the acceptability of implicit arguments in non-contrastive contexts, this effect is neutralized in fragments (Heidinger 2024). While in non-contrastive contexts, such as (5), a human antecedent would be degraded, human referents are possible as antecedents of null in contrastive contexts, as illustrated by (3). This neutralization of the animacy restriction in contrastive contexts suggests that the implicit argument is of a different nature in fragments (ellipsis / PF-deletion; see Merchant 2005) than in non-contrastive contexts (null pronoun).

An additional empirical argument for assuming different types of implicit arguments comes from Spanish: prepositions such as *con* 'with' and *sin* 'without', which do not allow for implicit arguments in non-contrastive contexts, allow them in contrastive ones (see Section 3.2 and Heidinger 2024).

We follow Heidinger (2024) in the assumption that depending on the syntactic-pragmatic context, syntactically projected implicit arguments fall in two different types: ellipsis in the case of a contrastively focused preposition, and a null pronoun in non-contrastive contexts. This gives us the threefold taxonomy introduced at the beginning of this section (see Figure 1).

Although we have presented the implicit argument in example (1b) as an example of a null pronoun (having a sloppy reading and no contrastive focus on the preposition *avec*), the parallelism of its structure might still suggest ellipsis. Still, the presence of a null pronoun has been clearly demonstrated since Zribi-Hertz (1984a,b) in the context of clitic left dislocation (henceforth CLLD) constructions. It is well known that in French, when a left-dislocated topic is co-referent with the logical object of the verb, a resumptive pronoun must occur in the comment, as shown in (4).⁷

- (4) *Cette valise_i, je l'_iai achetée l'année dernière.*
'This suitcase, I bought it last year.'
(Zribi-Hertz 1984a: 13; modified)

Drawing the parallel with left-dislocated topics co-referent with the logical argument of a preposition, Zribi-Hertz (1984a) points out that the same requirement must hold; (5) must also contain a resumptive pronoun co-referent with the topic.⁸

- (5) *Cette valise_i, je voyage toujours avec Ø_i.*
'This suitcase, I always travel with it.'
(Zribi-Hertz 1984a: 1; modified)

Further parallels can be made between CLLD involving verbal and prepositional complements. For example, bare indefinite quantifiers are infelicitous in CLLD constructions; the resumptive pronoun in both cases showing the same restrictions concerning the properties of the antecedent.

⁷ See De Cat (2005) for arguments that CLLD does not involve movement in French and similarly Zribi-Hertz (1984a) for a demonstration that the topicalized element does not raise in CLLD involving orphan prepositions. This contrasts with Italian, where it has been long claimed that CLLD does involve movement.

⁸ The presence of a resumptive clitic in the verbal domain, as in (4), and a null resumptive pronoun in the prepositional domain, as in (5), do not differ with respect to the status of the implicit argument, both minimally being ΦP following the analysis of Romance *l*-clitics as presented in Dechaine & Wiltschko (2002). As the verbal complement, the *l*-clitic can be assumed to raise to the inflectional domain, but as the complement of a preposition, raising out of an island is not possible.

- (6) a. **Quelque chose*_i, je l'_iai lu.
'Something, I read it.'
b. **Quelque chose*_i, je suis partie avec Ø_i.
'Something, I left with it.'

Beyond CLLD, Authier (2016) bolsters the argument for null *pro* with evidence of an implicit argument referentially dependent on a scope-bearing quantificational element, characteristic of pronouns. Hence, (7) can be interpreted such that for every occasion on which Marine brings home wild mushrooms, her husband does not cook with these mushrooms. In this case, the null pronoun acts as a donkey pronoun.

- (7) Quand Marine ramène [des champignons sauvages]_i,
son mari ne cuisine jamais avec Ø_i.
'Whenever Marine brings back wild mushrooms,
her husband never cooks with them.'
(Authier 2016: 240-241; modified)

Finally, we add that a syntactically projected implicit argument of *avec* – which can be neither a trace nor a result of ellipsis – is required to account for the gap in (8).

- (8) C'est [le genre de personne]_i que tu dois connaître t_i avant de parler avec Ø_i.
'He is the kind of person that you need to know before talking to.'

In summary, we assume that the implicit argument of French prepositions can take at least two and possibly three different shapes: it can be a null pronoun; it can be the result of ellipsis (as PF-deletion); and it may be, in some cases, an unprojected argument (following Authier's (2016) analysis of intransitive *pour*, for example). We limit the term *orphaning* / *orphan preposition* to cases where the implicit argument is a null pronoun.

3 Orphan prepositions

The term *orphan* in the context of prepositions was first coined in Postal (1970) and refers to the phenomenon that is currently called *preposition stranding* (the complement of the preposition raises as a *wh*-element and leaves behind an orphaned preposition).⁹ Zribi-Hertz (1984a,b) famously repurposes it in the term *orphan preposition*, which she uses to refer to prepositions with an implicit argument in French that are not a result of movement, as shown in (5) and repeated in (9) below.

- (9) Cette valise_i, je voyage toujours avec Ø_i.
'This suitcase, I always travel with it.'
(Zribi-Hertz 1984a: 1; modified)

French is noteworthy for the availability of such implicit arguments in contexts where they are sharply ungrammatical not only in English but also in closely related languages like Spanish.

- (10) a. *This suitcase_i, I always travel with Ø_i.
b. *Esta maleta_i, siempre viajo con Ø_i.

⁹ Stranding phenomena (e.g., *things which I am grateful for*; *Who did you go with?*) are associated with terms such as "hanging" or "dangling" preposition, atheoretical labels often found in grammars and style guides.

In what follows, we review the basic facts regarding lexical restrictions on orphaning. The situation in French has received considerably more attention, which is why we take it as a starting point, followed by a brief description of the situation in Spanish. It emerges that the only categoric difference involves certain simple prepositions, such as *avec* and *con*, exemplified in (9) and (10b). Hence, French is not as typologically special in having orphan prepositions (as it may seem).

3.1 French

The availability and distribution of orphan prepositions in French varies according to the prepositional element itself (see Zribi-Hertz 1984b: 58-59; Ilinski 2003: 269-270; Poplack et al. 2012: 218; Troberg 2020: 191-194; Tseng 2021: 791-792). *Par* 'by', for example, cannot be used as an orphan preposition.

- (11) *Paris_i, je suis passé par Ø_i.
 'Paris, I have passed through it.'
 (Zribi-Hertz 1984a: 11; modified)

Different accounts have been proposed as to which French prepositions can be orphaned (see Troberg 2020 for a recent overview). A distinction that figures prominently in this discussion is the one between *prépositions incolores* (e.g., *à* 'at', *de* 'of', *en* 'in') and *prépositions colorées* (*contre* 'against', *parmi* 'among', *vers* 'towards') with an intermediate category of *prépositions mixtes* (e.g., *avec* 'with', *dans* 'in', *pour* 'for', *par* 'by'), first proposed in Cadiot (1997) – a distinction which strongly relies on the semantic weight of the preposition. Despite capturing some tendencies with respect to the distribution of orphan prepositions, this tripartition has no predictive utility. The most robust generalization in this respect seems to be that *prépositions incolores* do not permit orphaning. As for the *prépositions colorées*, most of them seem to allow orphan uses; there are, however, exceptions such as *parmi* 'among' and *chez* 'at the home of' which would be considered *colorées* but do not allow orphaning. The intermediate class (*mixtes*) is the locus of most variation in French: While some allow orphaning (*pour*, *avec*, *sans*) others do not (*dans*, *sur*, *sous*, *par*). Quantitatively, the relevance of semantic weight for orphaning has been shown by Poplack et al. (2012: 219). In their corpus data of Canadian French, the percentage of orphaning is considerably higher for semantically strong prepositions than for weak ones (35% vs. 0.2%; Poplack et al. 2012: 219).

3.1.1 Decomposing P

While the meaning of the preposition clearly correlates in some way with the licensing of a null pronoun, the correlation is ever more obvious with complex SPEX involving nominal elements, such as *à côté (de)* 'beside', *en face (de)*. In order to align orphaning with both the compositional meaning of SPEX and their distribution, we adopt as our primary analytic tool an extended PP such as that developed in Svenonius (2010 et subsequ.). The syntax and semantics of spatial prepositions, simple or complex, can be decomposed under this approach into a hierarchy of multiple projections that may or may not have phonological content. The projections most important to our discussion are shown in Figure 2. Finer-grained analyses would require more projections, not shown here.

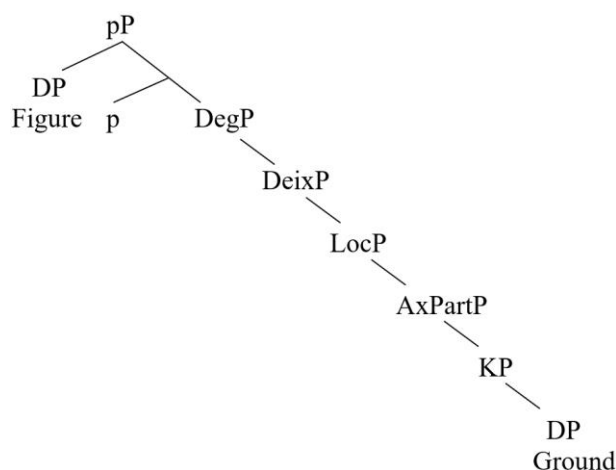


Figure 2: The extended PP following Svenonius (2010)

In what follows, we correlate the possibility of orphaning in French with patterns of lexicalization within the extended PP. Three basic types emerge: weak prepositions, strong prepositions, and axial part complexes.

3.1.2 Weak prepositions

The highest functional category, *p*, introduces the external argument, or Figure, and establishes a spatial relation between it and a Ground, the reference object, assigning case to the latter. This category encodes basic topological relations such as containment, contact, support, etc. and is the locus of semantically weak prepositions such as *à*, *de*, *en*, *sur*, *sous*, *dans*. In the cognitive semantics tradition, Vandeloise (1991) and especially Herskovits (1986) propose that such functional prepositions define geometric spaces along the three dimensions; for French, *à* 'at' and *de* 'of' define points, *sur* 'on' and *sous* 'under' define a two-dimensional space (a line or plane), and *en* 'in' and *dans* 'in' define spaces that must be three dimensional. The extended counterparts of *à* and *de* include *vers* 'toward' and *dès* 'away from', and *via* and *par* 'by, past, through' each define a point on a trajectory.

Crucially, *p*-elements do not license null pronouns in French, as illustrated in (11) for *par*. Troberg (2020), following Roberge (2012), attributes this restriction to their underspecified semantic content based on the same principle that underlies the absence of null objects with light verbs. Unlike *pro* subjects, which are widely taken to be licensed externally through inflection, null objects in French and Spanish are licensed internally by the semantic properties of the predicate, which, along with pragmatic inference, allow the null object to be recovered from the context; see Cummins & Roberge (2004, 2005) and Pérez-Leroux et al. (2008). While light functional verbs do not license null objects, lexical verbs do. Compare the examples in (12) for which (12a) shows the grammaticality of a null object with a lexical verb like *lire* 'read', while the null object of *faire*, a light verb, is ungrammatical, shown in (12b). As observed in Roberge (2012), this has a direct correlate in the prepositional domain. Strong prepositions permit orphaning (13a), but weak prepositions do not (13b).

- (12) a. "Tu as lu les pages_i?" [...] Il avait *lu* Ø_i.
 "'Did you read the pages?" He had read them.'
 (Larjavaara 2000: 43; modified)
- b. "Tu as fait tes devoirs_i?" *Il avait *fait* Ø_i.
 "'Did you do your homework?" He had done it.'

- (13) a. Le mur_i, Pierre était appuyé *contre* Ø_i.
'The wall, Pierre was leaning against it.'
(Zribi-Hertz 1984a: 19)
- b. *Mon pouvoir_i, Pierre est *en* Ø_i.
'My power, Pierre is in it.'
(Zribi-Hertz 1984a: 11)

To illustrate the relevance of semantic content, let us compare *dans* and *dedans* 'in' as a minimal pair denoting containment where only *dedans* allows orphaning. While *dans* establishes a basic relation of inclusion between a Figure and a Ground, *dedans* is more specific in that the referent of the Ground DP (i.e., the complement of the preposition) must have, in some sense, discrete boundaries. The uses of *dedans* are therefore a subset of the uses of *dans*, which illustrates the richer and thus more restrictive semantics of *dedans* (see Authier 2016: 251-259 and Troberg 2020: 197).¹⁰

- (14) a. L'avion est dans le ciel. *le ciel* = Ground, *l'avion* = Figure
'The plane is in the sky.'
- b. L'homme est dans la foule.
'The man is in the crowd.'
- c. La lettre est dans son enveloppe.
'The letter is in its envelop.'
- (15) a. *Le ciel_i, les oiseaux volaient dedans Ø_i.
'The sky, the birds fly inside it.'
- b. *?La foule_i, un homme a crié dedans Ø_i.
'The crowd, a man yelled inside it.'
- c. Cette eau_i, il y a du chlore dedans Ø_i.
'This water, there is chlorine in it.'

Distributionally, some weak prepositions can co-occur with other prepositional elements, most commonly as the head of complex SPEX such as *en face de* 'in front of/facing' or *par dessus de* 'over', in which the second element typically has nominal properties. But weak prepositions that denote a point of origin (*de*, *dès*) can also co-occur with more lexically contentful simple prepositions such as *avant* 'before', *après* 'after', *avec* 'with', and *entre* 'between'.

- (16) a. Cette aile date *d'avant* la transformation du prieuré en ferme.
'This wing dates from before the transformation of the priory into a farm.'
(Melis 2003: 16; modified)
- b. Il téléphonera *dès après* le repas.
'He will call after the meal.'
(Melis 2003: 35; modified)
- c. séparer l'ivraie *d'avec* le bon grain
'separate the chaff from the good seed (wheat)'

The structures in (17) illustrate the locative syntax of (14a), (16c), and that of *en face de la cathédrale* 'in front of/facing the cathedral'.¹¹ SPEX like *avec* and *face* lexicalize properties of

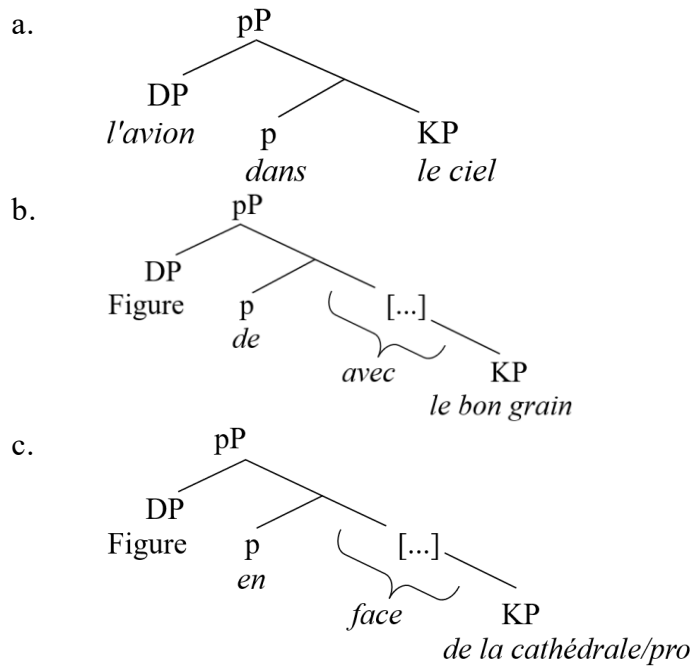
¹⁰ A native speaker points to an example such as (i) a possible counter example, but we argue that in so far as *le désert* 'the desert' is interpreted as a well-defined region inside of which cacti grow, then it should be acceptable.

(i) Les cactus, ils poussent dedans Ø. (Ø = le désert)
'Cacti grow in it (it = the desert)'

¹¹ The weak preposition *dans* in (17a) may be associated with additional features (not shown here), distinguishing it from *en* which would be strictly *p*. Crucially, however, we do not take the position that the decomposition of weak prepositions

the region associated with the Ground DP, instantiated by projections between *p* and KP, which we elaborate on in Sections 3.1.3 and 3.1.4. KP is a projection of Kase, a function that maps an entity (the denotation of the Ground DP) to its locus, or Eigenspace (Wunderlich 1991).

(17)



3.1.3 Strong prepositions

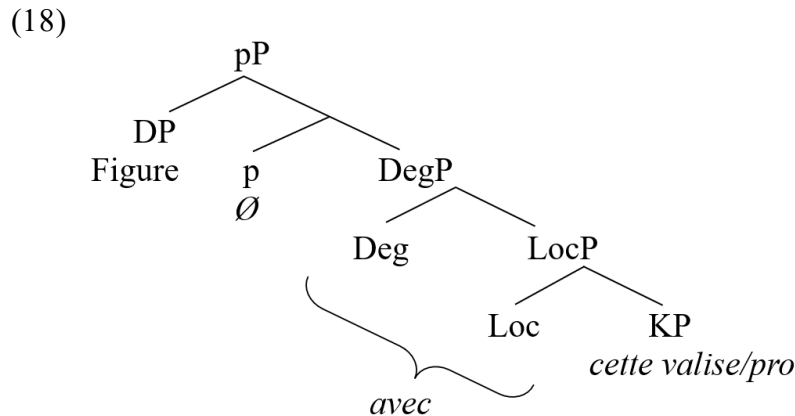
Prepositions like *avec* 'with', *entre* 'between', and *après* 'after' have more conceptual content than weak prepositions. They denote complex spatial relations between the Figure and the Ground that involve properties such as proximity (e.g., central coincidence for *avec*), whether the Ground entity is bifurcated (in the case of *entre*), and linear sequence (for prepositions like *après* and *avant*), among others. In French, these prepositions are morphologically simple, and, importantly, some members of this class license a null pronoun complement, shown for *avec* in examples (5), (7), and (8) and for *contre* in (13a). Following a cartographic approach, we assume that the descriptive content of such words is derived by association with heads between *p* and K. Strong prepositions lexicalize functional material closely associated with the meaning of the Ground DP and the region it occupies. For this reason, we call them "strong", by way of contrast with "weak" prepositions, which are the most functional elements, so to speak, of the extended projection, and which impose the fewest restrictions on the denotation of the Ground.

Apart from encoding a richer semantics than weak prepositions (*p* elements), strong prepositions have a distinct syntactic distribution: they may co-occur with *p* elements, shown in (16), but they never co-occur with axial part terms, such as *face* 'front' and *côté* 'side', while *p* elements do (e.g., **entre côté de* vs. *à côté de*). These two facts support a cartographic approach that, on the one hand, distinguishes strong prepositions from weak (since they can co-occur), and that, on the other, formalizes the overlapping functions of strong prepositions like *entre* and axial part terms like *côté*, accounting for their complementary distribution.

The tree in (18) shows the parallel structure of *avec* with both an overt DP and a null pronoun. As a bounded preposition denoting central coincidence, we propose that *avec*

mirrors complex prepositions with similar meanings. On distributional grounds, *dans* and *dedans*, for example, would not lexicalize the same set of functional heads, contrary to views expressed in Martin et al. (to appear) and Authier (2016).

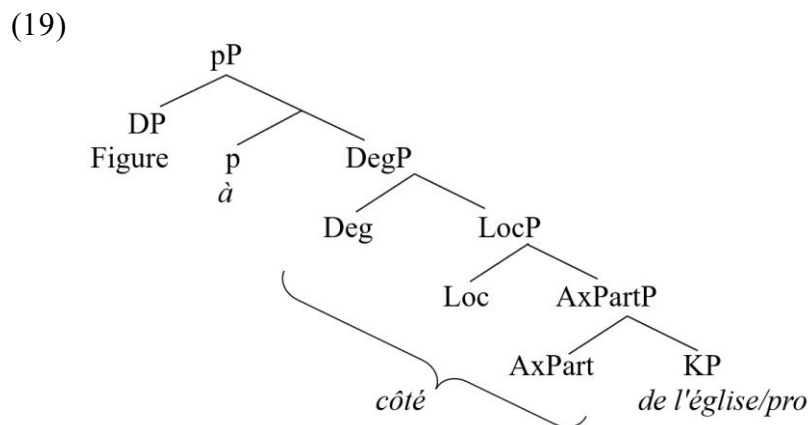
lexicalizes Loc[ation] carrying a non-projective feature and conceivably Deg[ree] encoding closeness to the Ground.



The licensing of the null pronoun would be related to the enriched semantic content of these elements, which, in this framework, corresponds to a richer syntactic structure; compare with (17a).

3.1.4 Axial part complexes

Morphologically complex SPEX such as *autour (de)* 'around', *à côté (de)* 'beside', *en face (de)* 'in front/facing', *au pied (de)* 'at the foot', are superficially composed of two parts: a weak preposition and a morpheme, often nominal, that denotes an axial part, a part of an object that is determined by its natural axes: front, back, top, bottom, side, circumference, etc. (Jackendoff 1996). These SPEX often have projective semantics in the sense that, for example, *à côté* 'beside' does not denote the side of an object but rather the region of space that projects outward from its side, a meaning derived by a dedicated functor, Loc, as proposed in Svenonius (2010) following Zwarts (1997) and Zwarts & Winter (2000) for the creation of vector spaces. The parallel structure for *à côté de l'église* 'next to/beside the church' and its orphan equivalent is given in (19).



We discuss the nature of French axial part terms in Section 4.1.2, which bears directly on the far less restrictive licensing of null pronouns with these complexes in French, Spanish and other languages.

3.2 Spanish

Although there appears to be an almost complete absence of comparative work concerning orphan prepositions, Zribi-Hertz (1984a) observes that Spanish (see (10b) and (20b)) and Brazilian Portuguese, two closely related languages to French, do not allow orphaning to the same extent as French.

- (20) a. Mi sombrero_i, vine con él_i.
'My hat, I came with it.'
b. *Mi sombrero_i, vine con Ø_i.
(Zribi-Hertz 1984a: 19; modified)

Campos (1991: 746) asserts that orphaning is possible in Spanish at least with a limited set of prepositions (e.g., *encima* 'on top of' in (21); see also Pavón Lucero 2003, Fábregas 2007, Fraga 2017, 2020).¹²

- (21) A: ¿Qué puso encima del estante_i?
'What did she put on top of the shelf?'
B: Puso libros encima Ø_i.
'She put (some) books on top.'
(Campos 1991: 746; modified)

Zribi-Hertz (1984a: 21-22) is aware of such uses but considers them a different phenomenon than orphaning. Rather, she assumes that the alternation between an implicit argument (a *gap* in her terms) and an overt pronoun, as in (22), is stylistic in the sense that there is no evidence of a semantic difference between the two (e.g., with respect to animacy restrictions). The implicit argument would simply belong to a more informal register.¹³

- (22) a. La casa_i, Pedro estaba delante Ø_i.
b. La casa_i, Pedro estaba delante de ella_i.
'The house, Pedro was in front (of it).'
(Zribi-Hertz 1984a: 21-22; modified)

Notably, Campos (1991) gives no examples of orphaning for *con* 'with' and *sin* 'without', the equivalents of the prototypical and well attested orphan prepositions in French. In an acceptability experiment, Heidinger (2024) shows for these two prepositions that orphaning is ungrammatical in non-contrastive contexts. But just like in French, we also see an effect of the syntactic-pragmatic context on the availability of implicit arguments in Spanish: implicit arguments are significantly more acceptable in fragments with contrastive focus on the preposition, as in (23B) (see also Pavón Lucero 2023: 471).

¹² SPEX like *encima* 'on top of', *delante* 'before', *detrás* 'behind' etc. are not uniformly called *prepositions* in the literature. Especially in traditional grammars, where having an overt complement is a defining criterion of prepositions, these expressions would not be considered prepositions (see Pavón Lucero (2003: 125-126) for a discussion and Section 4.2.2 for arguments that these expressions do have a projected implicit argument).

¹³ In Zribi-Hertz (1986: 624-629), such cases seem to be analysed on par with French and are thus considered to be orphan prepositions.

- (23) A: ¿Corres más rápido con los zapatos_i?
 'Do you run faster with the shoes?'
 B: No, *sin* Ø_i.
 'No, without.'

This asymmetry in acceptability aligns with the results of preliminary corpus searches for Spanish *con* and *sin* with implicit arguments.¹⁴ They appear to be much more frequent in contrastive contexts as shown in (23) and (24) than in non-contrastive ones; cases like (25) without clear overt contrast are indeed hard to find.

- (24) A: ¿Con o sin guarnición_i?
 'With or without garnish?'
 B: Como no soy militar, *sin* Ø_i.
 'As I am not in the military, without.'
 (Cabrera Infante, 1995, *Una mujer que se ahoga*; CREA, modified)
- (25) Hice una procesión de farmacia en farmacia hasta que en una se creyeron la historia de que se me había olvidado la receta_i y me las dieron *sin* Ø_i.
 'I proceeded made a procession from pharmacy to pharmacy until one of them believed the story that I had forgotten my prescription, and they gave them (= pills) to me without it.'
 (Etxebarria, 2001, *De Todo lo Visible y lo Invisible*; CORPES, modified)

Troberg (2020: 203) notes that, like French, orphaning in Spanish is possible with complex prepositions (*locutions prépositives* in her terms), but unlike French, orphaning is ungrammatical with simple prepositions. Indeed, most of the prepositions given in Campos (1991) which allow orphaning in Spanish are complex prepositions in the sense that both a *p* element and an axial part term are overtly present (*encima* 'on top', *enfrente* 'in front of', etc.). Nevertheless, the simple preposition *para* seems to permit an implicit argument in non-contrastive contexts at least for some speakers, shown in (26), suggesting an orphaning analysis, but ellipsis shouldn't be ruled out in this context either.

- (26) A: ¿José trabaja para ese señor_i?
 'Does José work for this man?'
 B: Sí, trabaja *para* Ø_i.
 'Yes, he works for him.'
 (Campos 1991: 743; modified)

Terminologically, Campos (1991) distinguishes between "regular" and "substantive" prepositions, where only the latter allow for orphaning. A correlate of this distinction, and hence of the availability of orphaning, mentioned by Campos is the necessity to introduce overt complements with the preposition *de*. The class of Spanish SPEX that permit orphaning therefore also includes *cerca (de)* 'close', *dentro (de)* 'inside', *fuera (de)* 'outside', etc. for which the presence of a light *p* is not evident. Again, non-spatial *para* is an apparent exception in that it does not need *de* but still can appear with an implicit argument (yet it is presented by Campos (1991: 743) as a substantive preposition).

To close this brief overview, it must be mentioned that animacy constraints do not seem to hold in Spanish to the same extent as they do in French. Neither for implicit arguments, nor

¹⁴ We searched in CREA and CORPES for the prepositions *con* and *sin* followed by a full stop (to exclude cases with an overt complement).

for strong pronouns. Heidinger's (2024) experimental data show that strong pronouns as complements receive very high acceptability scores independently of whether they refer to human or to non-human referents. This is in line with Jensen's (1973: 118) corpus findings, where strong pronouns in Spanish PPs refer about equally often to human and non-human referents (see (27) for the latter). It is also in line with Fernández Soriano's (1999: 1220), Alonso Ovalle & D'Introno's (2000) and Pomino's (2017: 702) statements that as complements of prepositions, strong pronouns are not restricted to humans.

- (27) La verdad_i es el arma más potente de la justicia. *Sin ella_i*, no habría justicia.
 'Truth is the most potent weapon of justice. Without it, there would be no justice.'
 (Jensen 1973: 120; modified)

Finally, null pronouns (i.e., implicit arguments in non-contrastive contexts) receive much lower acceptability scores than strong pronouns in Spanish (1.68 vs. 4.62), and as with strong pronouns the acceptability does not depend on the humanness of the referent (Heidinger 2024: 23, Appendix). While Heidinger (2024) only tested *sin* and *con*, Campos's (1991) example in (26) with an implicit argument referring to a human further highlights the absence of animacy restrictions.¹⁵

4 SPEX

4.1 French

4.1.1 Simple prepositions: weak and strong

Weak prepositions, which we take to lexicalize little *p*, do not permit a null pronominal complement. These include the four quintessentially functional prepositions, *à* 'at', *de* 'of', *en* 'in', and *par* 'by', along with other weak prepositions that denote basic topological relations and whose meaning has wide uses, both spatial and notional: *sur* 'on', *sous* 'under', *dans* 'in', *hors de* 'out of', *vers* 'toward', *dès* 'from', and *via* 'via'.

In contrast, strong prepositions encode more complex relations between a Figure and a region associated with a Ground. We take them to minimally lexicalize Loc and Deg. These elements are not nominal, and of those with a spatial meaning, several license a null pronoun: *avec*, *contre*, *entre*, and *après*.¹⁶ While the licensing of a null pronoun is on the one hand categorically excluded from *p*-elements and on the other categorically available with axial part terms, orphaning is idiosyncratic with strong prepositions.

Like *avec*, the antecedent of the implicit arguments of *après*, *contre* and *entre* can occur as a left-dislocated topic (shown in the (a) examples in (28-29)), a strong argument for a null pronoun analysis. In addition, the implicit argument for each can be referentially dependent on a preceding quantificational element (shown in the (b) examples) just as an overt pronoun would. Our native speaker informants find the examples in (28-29) to be perfectly acceptable, but some are felt to belong to informal registers.

¹⁵ The judgments of a native speaker reviewer suggest, however, interspeaker variation as concerns the effect of animacy on the acceptability of implicit arguments in Spanish PPs.

¹⁶ We consider *avec* to have spatial semantics given that, in a subset of its uses, it requires proximity. Inferences of possession, accompaniment, simultaneity, part-whole relations, manner/means, etc. would be secondary in such cases. *Sans*, on the other hand, is less clearly spatial, and is associated with the notions of (not) having and (not) being. See Choi-Jonin (2006) and references therein.

- (28) a. Son argent_i, Pierre court toujours après Ø_i.
'His money, Pierre is always chasing after it.'
- b. Quand mon chien voit des voitures_i passer, il aboie toujours après Ø_i.
'When my dog sees cars going by, he always barks after them.'
- (29) a. Ces deux cyprès_i, il faut passer entre Ø_i sans les toucher.
'These two cypress trees, you need to pass between them without touching them.'
- b. Lorsque Marie met [deux vases de marguerites]_i sur la table, Jean met toujours un vase de roses entre Ø_i.
'When Marie puts two vases of daisies on the table, Jean always puts a vase of roses between them.'

Depuis does not permit orphaning in its locative sense designating a point of origin, but in its temporal meaning of 'since' it does. Other strong prepositions allow null pronouns as well, but they do not have spatial semantics: *pour* 'for', mentioned earlier, *sans* 'without', and *selon* 'according to'.

The licensing of a null pronoun is also subject to change. For example, *avec* 'with' and *parmi* 'among' are both attested with implicit arguments in medieval French (30), but while *avec* continues to be robustly used in orphaning contexts, *parmi* is not, no longer having its axial part structure or its meaning of 'through the middle'.

- (30) a. Et li roi_i [...] entra enz a privee mesnie, et Artus son neveu avec Ø_i.
'And the king went in accompanied by a small party, and Arthur his nephew with him.' (*Récit d'un ménestrel de Reims*, 31v; BFM ; modified)
- b. [La chité de Nantes]_i est grande, et la riviere de Loire qui court parmi Ø_i, moult large.
'The city of Nantes is large, and the Loire river that runs through the middle, very wide.' (Froissart, *Chroniques D*, 495; DMF; modified)

Chez 'at the home of' is an interesting case of a strong preposition that never allows an implicit argument of any kind. The requirement to have only overt DP complements likely arose as a result of its reanalysis from a construct state noun, which was always followed by an overt possessor DP (see Longobardi 2001). This former distributional property would have become a lexical restriction in its life as a preposition.

4.1.2 Axial part complexes

The null pronoun as a complement in axial part complexes has far fewer restrictions than that of strong prepositions; temporal, and notional extensions of spatial meanings tend to be just as acceptable, and they are, in general, register neutral.

Tests for the pronominal status of the implicit argument are convincing; its antecedent can be a left dislocated topic (31a, 32a); it can be referentially dependent on a preceding quantificational element (31b, 32b); and it allows sloppy readings (31c, 32c).

- (31) a. Cette table_i, j'aimerais m'asseoir juste à côté Ø_i.
'I'd like to sit right beside this table.'
- b. Quand Marie achète un nouveau lit_i, son mari met deux tables de chevet à côté Ø_i.
'Each time Marie buys a new bed, her husband puts two bedside tables beside it.'
= For each new bed that Marie buys, her husband puts two night tables beside it.
- c. Marie_i a mis trois livres sur son_i lit et Jean_j en a mis deux à côté Ø.
'Marie put three books on her bed and Jean put two beside (it/his).'
- Ø = Marie's bed or Jean's bed
- (32) a. Cette montagne_i, le trésor est enfoui en dessous Ø_i.¹⁷
'The treasure is buried underneath this mountain.'
- b. Quand Jean sort une nouvelle nappe_i, sa femme cache les serviettes en dessous Ø_i.
'Each time that Jean puts out a new tablecloth, his wife hides the napkins underneath.'
= For each new tablecloth that Jean puts out, his wife hides napkins underneath it.
- c. Marie_i place trois fleurs sur son_i bureau, tandis que Jean_j en place deux en dessous Ø.
'Marie puts three flowers on her desk while Jean puts two underneath.'
- Ø = Marie's desk or Jean's desk.

The question of the category of the axial part nominal naturally arises. Roy (2006) and Roy & Svenonius (2009) treat it as functional, but cross-linguistic approaches to such SPEX commonly use the term "relational noun", and there is compelling evidence that the axial part component of these SPEX in French is lexical. Matushansky & Zwarts (2019) present evidence that the central element of axial part SPEX in French and various other languages displays the characteristic properties of weakly referential nominals. Real Puigdollers (2021) offers similar arguments for Central Catalan. Like nouns in French, axial part terms can encode definiteness, number, and gender in the article, and they also have full nominal counterparts with similar semantics and morphosyntax, including the requirement that their complement is genitive case marked by *de* 'of' and the possibility of a null pronoun as a complement.¹⁸

- (33) a. Nous vivons à côté (de l'église).
'We live beside {it/the church}.'
- b. Nous vivons du côté ouest (de l'église).
'We live on the west side (of the church).'

¹⁷ One of our consultants prefers this sentence to include the locative clitic pronoun *y*, as in (i). See Authier (2016: 244-248) for a brief discussion of the co-occurrence of *y* with axial part terms when the implicit argument is inanimate and clearly locative. There is much more to say about this phenomenon, both in terms of its syntax and its value as a diagnostic, which space limitations do not allow us to develop here.

(i) Cette montagne, le trésor *y* est enfoui en dessous.
'This mountain, the treasure is buried underneath there'

¹⁸ Some axial part nominals like *devant* and *derrière* do not occur with the typical elements of an axial part complex, that is, with neither an outer weak preposition nor a complement introduced by genitive *de*. We assume they have the same syntactic structure as other axial part complexes, but that the axial part incorporates into *p* and *p* itself licenses and assigns case to the embedded DP. See also Real Puigdollers (2021) for discussion of similar cases in Central Catalan. In all other respects, *devant* and *derrière* behave like projective SPEX.

Matushanksky & Zwarts (2019) further point out that axial parts pattern like weakly referential nominals like *bed* (*Pierre va au lit* 'Pierre is going to bed') and *school* (*Pierre est au lycée* 'Pierre is at school') in that they are neutral in terms of number (34), and do not allow adjectival modification (35).

- (34) a. Les gamins vont {au lit/*aux lits} de bonne heure.
'The kids go to bed/*beds early.'
- b. {au pied/*aux pieds} de la montagne
'at the {foot/*feet} of the mountain'
- (35) a. Les gamins vont au (*petit) lit de bonne heure.
'The kids go to (*small) bed early.'
- b. au (*grand)pied de la montagne
'at the (*big) foot of the mountain'

In this light, the real Ground reference object of such SPEX includes both the axial noun and the embedded genitive-marked DP with which it holds a part-whole relationship (e.g. *le pied de la montagne*). Null pronouns as complements of these SPEX would not be licensed by a preposition per se, but rather by the axial nominal; in other words, the implicit argument of the axial part nominal *côté* in (33a) is licensed in the same way as the implicit argument of the noun *côté* in (33b). Axial nominals presuppose the existence of a whole, which can be left implicit even in languages that do not otherwise permit orphaning. This is demonstrated, for example, in Ursini & Huang (2020) for Mandarin and is shown for English in Svenonius (2010: 136-141). This basic difference between the licensing of nominal complements denoting wholes and prepositional complements which do not, could be conceived in structural terms along the lines of Fraga (2020): genitive-marked arguments are merged as the specifier of a functional projection, whereas non-genitive marked arguments are true complements of the preposition.¹⁹

4.1.3 Obligate orphans

A special case of axial part complexes are those that only ever occur with an implicit argument. *Dessous* 'underneath', *dessus* 'on top', *dedans* 'inside', and *dehors* 'outside', often called *spatial adverbs*, do not license overt complements. For example, in contemporary standardized French, example (36a) would be ungrammatical with an overt complement. However, as a noun (36b) and within an axial part complex (36c), an overt complement is licensed.

- (36) a. Le chat dort *dessus* {Ø/*de l'étagère}.
'The cat sleeps on top (of the shelf).'
- b. Le chat dort sur *le dessus de l'étagère*.
'The cat sleeps on the top of the shelf.'
- c. Elle a accroché le tableau *au-dessus de son lit*.
'She hung the picture above her bed.'

That the antecedent of the implicit argument of these four forms can be a left-dislocated topic is a strong argument in favour of a null pronoun analysis, as shown in (15c) for *dedans*. In addition, the implicit argument can have a full range of interpretations shown in (37) for one and the same SPEX, *dedans*: definite (37a); dependent on a preceding quantificational element

¹⁹ Troberg (2020) proposes that in French genitive-marked arguments merge as the complement of the head. The null allomorph of genitive *de* occurs in *pro* contexts yielding a null genitive KP (e.g., *à côté Ø_{de} pro*). This may also be the case for Spanish, or alternatively, the null genitive argument may be a member of the genitive pronoun paradigm (*suyo*, etc.).

(37b); deictic proximal 'here' (37c); and generic in the sense that any appropriate interior space is licit in (37d).

- (37) a. Il ouvre la boîte_i et regarde dedans Ø_i.
'He opens the box and looks inside it.'
- b. Quand mon chat voit [une boîte en carton]_i, il essaie de se cacher dedans Ø_i.
'Whenever my cat sees a cardboard box, he tries to hide inside it.'
- c. Mets les livres dedans Ø. (pointing to a box)
'Put the books inside.'
- d. Quand il pleut, les chats préfèrent être dedans Ø. (Authier 2016: 257)
'When it rains, cats prefer to be inside.'

The interpretation of the null pronoun of *dehors* 'outside' is more restricted, however. Its interpretation is typically generic in the sense of 'outside' any context-appropriate building (as in (38a)). This restriction gives rise to the strange interpretation in (38b).

- (38) a. Parlons en *dehors* Ø.
'Let's talk about it outside.'
≠ 'Let's talk in the hallway, outside the room we are currently standing in'.
- b. Quand on met la grenouille dans la boîte, elle saute tout de suite *dehors* Ø.
'When we put the frog in the box, it immediately jumps outdoors.'
≠ 'When we put the frog in the box, it immediately jumps out (of the box).'

It appears then, that as axial prepositions, *dessous* 'underneath', *dessus* 'on top', *dedans* 'inside', and *dehors* 'outside' are obligate orphans in the sense that they are restricted to the function of orphaning; they only select *pro*.²⁰ There is a plausible historical explanation for this; *dessous* 'underneath', *dessus* 'on top', *dedans* 'inside', and *dehors* 'outside' could be used both transitively and intransitively in medieval French, and so their function overlapped considerably with their weak counterparts: *sous* 'under', *sur* 'on', *dans* 'in', and *hors/hors de* 'out (of)'. The principal difference between them was that the weak prepositions were never used intransitively. These near doublets became functionally distinct early on so that *de*-forms were rarely used in transitive contexts; see Rainsford (2019a,b) for a diachronic description. This situation gave rise to the *de*-forms being acquired with a radically restricted selectional requirement: only *pro*. In support of this view, Martin et al. (to appear) point out that *derrière* and *devant*, which do not have simple counterparts, are consequently not reanalysed as obligate orphans. We discuss these prepositions further in 4.2.3 in relation to the very similar Spanish SPEX *adentro* 'inside', *afuera* 'outside', *arriba* 'above', *abajo* 'underneath', *alante* 'in front', *atrás* 'behind'.

4.2 Spanish

Spanish SPEX fall into three classes depending on whether or not they allow overt and implicit arguments in non-contrastive contexts: (i) SPEX which only license overt arguments (e.g., *sobre* 'on', *bajo* 'under', *entre* 'between', *ante* 'before'), (ii) SPEX which only license implicit arguments (e.g., *arriba* 'on top', *abajo* 'below', *atrás* 'behind', *alante* 'in front'),²¹ and (iii) SPEX which can occur with overt and implicit arguments (e.g., *encima (de)* 'on top (of)', *debajo (de)* 'underneath', *detrás (de)* 'behind (of)', *delante (de)* 'in front (of)').

²⁰ As nouns, on the other hand, they select genitive complements as expected, and as incorporated elements into an overtly expressed *p* (*de dessous*, *par-dessus*, etc.), KP is assigned accusative case by *p*, as suggested for *derrière* and *devant*.

²¹ There is dialectal variation concerning this class in that *arriba*, *abajo*, *atrás* can be used with overt complements in non-European varieties (Pavón Lucero 2003: 89, 107; see also Fábregas 2007).

These distributional properties correlate to a certain extent with morphological properties, as shown by the series in (39) (also *ante*, *alante*, *delante* / *bajo*, *abajo*, *debajo*). An important generalization (at least for European Spanish) is that SPEX prefixed with *a-* only allow for implicit arguments (see Fábregas 2007: 2).

- | | | |
|---------|------------------------------|----------------------------|
| (39) a. | <i>tras</i> *(la puerta) | only overt argument |
| b. | <i>atrás</i> (*de la puerta) | only implicit argument |
| c. | <i>detrás</i> (de la puerta) | overt or implicit argument |
| | 'behind (the door)' | |

Restrictions on the complement types also correlate with differences in case assignment (Pavón Lucero 2003, Fábregas 2007). While SPEX which require an overt complement directly assign oblique case to their complement (*tras mí* in (40)), SPEX which can be used with or without overt complements need *de* to assign oblique case (*detrás de mí* in (41)) and can also directly assign genitive case to pronouns (42).²²

- (40) Escuchaba *tras mí* las ruedas de su carrito avanzando rápidas y suaves sobre la alfombra del vestíbulo.
'I could hear behind me the wheels of his cart rolling quickly and smoothly over the carpet in the hallway.'
(Salisachs, 1975, *La gangrena* (ESPAÑA); CREA, modified)
- (41) [...] para que mi madre esté *detrás de mí*.
'for my mother to be behind me'
(*Tiempo*, 30/07/1990; CREA, modified)
- (42) El libro está *delante suyo*.
'The book is in front of him.'
(Fábregas 2007: 4; modified)

4.2.1 Simple prepositions: weak and strong

In terms of the threefold typology introduced in Section 3.1 for French, all three types of SPEX are evident in Spanish as well. As a starting point, consider the following list of prepositions from the RAE & ASALE (2009) (this list is intended to be exhaustive, but one must bear in mind that one of the defining criteria for *preposition* in RAE & ASALE (2009) is that the complement cannot be omitted).

- (43) *a* 'at', *ante* 'before', *bajo* 'under', *cabe* 'near', *con* 'with', *contra* 'against', *de* 'from', *desde* 'from, since', *durante* 'during', *en* 'in', *entre* 'between', *hacia* 'toward', *hasta* 'until/up to', *mediante* 'through', *para* 'for', *por* 'by', *según* 'according to', *sin* 'without', *so* 'under', *sobre* 'over/on', *tras* 'after/behind', *versus* 'versus', *vía* 'via' (RAE & ASALE 2009: §29.2(a))

Some of the SPEX in this list can be identified rather easily as pertaining to the class of weak prepositions due to their reduced semantic content and extended functions. On these grounds, *a*, *de*, *en*, and *por* are prototypical examples. Distributionally, they also occur as the outer preposition of complex SPEX, along with *desde* and *para*, and so we therefore take them as lexicalizing the highest head *p* in the extended PP. In addition, Fraga (2020) distinguishes *ante*,

²² Searches for *tras mí* and *detrás de mí* in CREA yield very different numbers of hits: 4 vs. 200; this difference does not follow from a general difference in frequency since *tras* is about four times as frequent as *detrás* in the corpus. Instead, it is most probably due to different animacy restrictions of the SPEX themselves (see also Fraga 2017: 108-109)

bajo, *sobre*, and *tras* from their *de*- counterparts, showing that they permit bare noun arguments (while the *de*- series do not) and that they do not license two coordinated DPs (while the *de*-series do). If these two diagnostics identify them as members of the little *p* class, it would be expected that none of them license *pro*, and French and Spanish would converge on the core elements of weak prepositions.

While the differences between French and Spanish in terms of orphaning tend to be highlighted, they seem to converge on more points than they differ. A comparative approach reveals that, among strong prepositions (i.e., morphologically simple, non-nominal, assigning oblique case), the only differences between French and Spanish reside in four strong prepositions having spatial semantics (*con* 'with', *contra* 'against', *entre* 'between', and *tras* 'behind') and two that do not designate clear spatial relations (*según* 'according to', *sin* 'without').²³ The six French equivalents of these allow orphaning.

Why do these Spanish prepositions disallow orphaning while the French equivalents allow it? Are these historical accidents? This may appear to be the case for *avec* < *apud hoc* 'near that', *après* < *ad pressum* 'at (what is) pressed', 'near', and *sans* < *sine* 'without' + *absens* 'remote, absent'; the French simple forms may still carry features of the complex spatial semantics they once had in Latin (as opposed to Spanish *con* < *cum*, *tras* < *trans*, and *sin* < *sine*), but the resistance of *parmi* 'among' < *par mi* 'through the middle' to orphaning would then require explanation as would the divergent behaviour of the cognates *entre/entre*, *contre/contra*, *selon/según*. The central question of what underlies this variation remains open.²⁴

4.2.2 Axial part complexes

Generally speaking, the equivalents of French and Spanish complex prepositions containing axial terms converge in that they permit implicit arguments: *enfrente* Ø/ *en face* Ø; *detrás* Ø/ *derrière* Ø; *debajo* Ø/ *en dessous* Ø, etc. Just as their French counterparts, these Spanish SPEX may receive a definite interpretation (referring to contextually given antecedents), a deictic interpretation,²⁵ or a generic non-referential interpretation (see (44-46)).

- (44) Conozco muy bien ese parque_i porque vivo justo *enfrente* Ø_i.
'I know that park very well because I live just live right across the street.'
(Pavón Lucero 2003: 137; modified)
- (45) A: ¿Dónde vive María? 'Where does María live?'
B: *Debajo* Ø. 'Below/under.'
(Ø = from me, from here)
(Pavón Lucero 2003: 137; modified)
- (46) *Desde fuera*, todo parece muy fácil.
'From outside, everything looks very easy.'
(Pavón Lucero 2003: 135)

²³ Recall that both languages allow implicit arguments with the non-spatial prepositions *pour* and *para* 'for' (although the orphaning of *para* is judged ungrammatical by a native speaker reviewer).

²⁴ Svenonius's (2010) account of spatial prepositions that license null *pro* in English requires that they lexicalize Loc, nicely accounting for their stronger semantic properties. This requirement rules out weak prepositions from orphaning along with any strong prepositions that raise to a position higher than Deix – namely Deg and *p* – since, according to Svenonius, the licensing of null *pro* requires movement of LocP to the specifier of DeixP. Assuming all bounded prepositions in English (e.g., *between*, *beside*, *among*, etc.) raise to Deg or *p*, they consequently do not permit orphaning. It remains an open question, however, if such an approach might capture the variation in French and Spanish (*avec* and *entre*, for example, lexicalizing Loc but not *p* in French, while *con* and *entre* would be exponents of *p* in Spanish). The kind of empirical arguments proposed for English are not as clear for Romance.

²⁵ This interpretation is available according to Pavón Lucero (2003: 137), whereas Fábregas (2007: 21) holds that the *de*-forms always have a non-deictic interpretation.

In the case of SPEX which allow for both overt and implicit arguments, Pavón Lucero (2003) claims that while the implicit argument is *pro* in anaphoric and deictic uses (Pavón Lucero 2003: 140), it is only semantically present in generic and non-referential uses (see (46) above). On the other hand, Campos (1991: 747) argues that such implicit arguments are a silent Topic operator which moves out of the complement position to a topic position at LF. We point out that the implicit argument of axial part complexes passes the diagnostics used by Authier (2016; see Section 2) for syntactically projected implicit arguments. Firstly, the implicit argument can be dependent on a higher quantificational element, typical of pronouns. Hence, according to our informants (47) can receive the interpretation that when María buys table₁, her husband will put three vases on top of table₁, when María buys table₂, her husband will put three vases on top of table₂ etc. Besides *encima* 'on top' similar sentences were tested for *debajo* 'under' and *detrás* 'behind' with the same result.

- (47) Cada vez que María compra una mesa nueva_i, su marido pone tres floreros *encima* Ø_i.
'Every time Maria buys a new table, her husband puts three vases on top of it.'

The second diagnostic is the availability of sloppy readings, another property of pronouns. In (48), the implicit argument of *detrás* may refer to Juan's body and is thus not co-referential with its formal antecedent (María's body); similar results were obtained for *debajo* and *encima*.

- (48) María_i cierra las manos delante de su_i cuerpo y Juan_j las cierra *detrás* Ø.
'Mary closes her hands in front of her body and John closes them behind {her body/his body}.'

Spanish has a large set of axial part complexes that are transparently built on a combination of light *p* and a noun, such as *al lado* (*de*), *en medio* (*de*), *enfrente* (*de*), etc., which also allow orphaning, just as French.

4.2.3 The *a*-series

Many peninsular varieties of Spanish have a morphologically distinct set of deictic SPEX that only select implicit arguments: *atrás* 'behind', *alante* 'in front', *abajo* 'under', *arriba* 'above', *afuera* 'outside', *adentro* 'inside'. As argued in Fábregas (2007: 20-22), the implicit argument of the *a*-series is *pro*, and the deictic interpretation is based on the position of the observer/speaker. *Pro* can be definite, picking its referent from the preceding context (49a); its reference can depend on a preceding quantificational element, expected of null pronouns (49b); the reference can also be established contextually, where proximal 'here' is the unmarked interpretation (49c); and the reading can be generic where any appropriate interior space is licit. A house is most salient in (49d).

- (49) a. Abrí la caja_i y miré *adentro* Ø_i.
'He opened the box and looked inside.'
b. Cada vez que abre una caja_i, mira *adentro* Ø_i.
'Each time that he opens a box, he looks inside.'
c. Pon los libros *adentro* Ø.
'Put the books inside.' (pointing to a box)
d. Cuando llueve, los gatos prefieren estar *adentro* Ø.
'When it rains, cats prefer to be inside.'

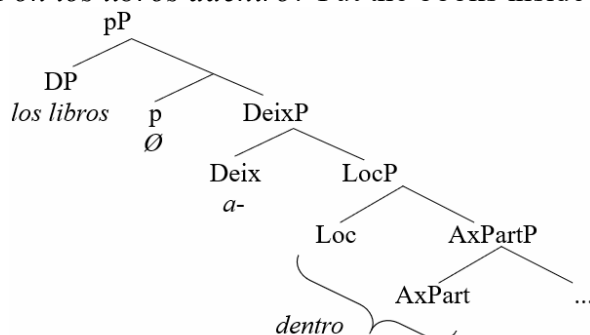
Compared to the Spanish *a*-series, the four French obligate orphans discussed in 4.1.3 (*dessous* 'underneath', *dessus* 'on top', *dedans* 'inside', and *dehors* 'outside') appear to pattern identically – compare the examples in (49) with those in (37). Beyond these similarities, we suggest that the morphology of the *a*-series sheds light both on the restriction of these elements in both languages to select null pronouns as complements and on the expression of deixis in the prepositional domain.

The category of the prefix *a*- is taken to be a locative preposition in Fábregas (2007) and a disjoint functor in Romeu (2014), both of which would roughly correspond to *p* in our framework. Notice in (50), however, that *de* can co-occur with *a*-series forms like *abajo*, and it always occurs to the left. This should not be possible under the assumption that *a* and *de* are both of the category *p* (see 4.2.1).²⁶ *A*- must therefore lexicalize a head lower in the structure.

- (50) Los deliciosos olores vienen *de abajo*.
'The delicious smells are coming from below.'

On the basis of its deictic semantics and its distribution with respect to *de*, we take *a*- to be minimally associated with Deix, a functional head dominating Loc but lower than *p* and Deg, following Svenonius (2010) and shown in (51).²⁷

- (51) *Pon los libros adentro*. 'Put the books inside'



The connection between Deix and *pro* in the prepositional domain can now be formalized drawing on Harbour's (2016) treatment of person and deixis within ϕ P. Harbour observes a wealth of evidence that person categories correlate with spatial partitions. Languages show many examples of morphological overlap; for example, the *me/here* alternation in (52a) and the diachronic reanalysis of spatial deictic expressions in Italian meaning 'here' and 'there' as personal pronouns 1PL and 2PL respectively in (52b,c).

- (52) a. Give it me/here.
b. *ci* (Italian) 'we' < *hince* (Latin) 'from here'
c. *vi* (Italian) 'you (pl.)' < *ibi* (Latin) 'there'

To capture this relation, Harbour proposes that within ϕ P, a deixis functor χ selects Person. We suggest that this could be applied to Spanish such that *a*-, [$\pm$ author] in Harbour's terms, lexicalizes the deictic and person sub-features of ϕ . ϕ P would then raise, as a clitic, to check an unvalued χ feature in Deix. Double marking of spatial deixis, such as the example in (53),

²⁶ *Por* as in *por abajo* 'below' is a further example of the co-occurrence of *a*- with a *p*-element.

²⁷ This prefix would be distinct from the homophonous weak preposition *a* in other SPEX (e.g., *al lado (de)* 'beside', *a la izquierda (de)* 'to the left of').

can be accounted for assuming phrases like *ahí* 'there' are merged, after clitic raising, as the specifier of Deix.

- (53) Pon los libros *ahí* a_i =dentro t_i .
'Put the books inside there.'

We further suggest that a similar analysis can be fruitfully extended to French obligate orphans. *Dedans*, *dessous*, and *dessus* commonly occur with the deictic proclitics *ci* and *là*, which lexicalize spatial deixis in combination with person: *ci* lexicalizes the space associated with first person (+author in Harbour's terms); and *là* is the instantiation of –author deixis. These clitics would raise, just as Spanish *a-*, to check an unvalued χ feature.²⁸ In the absence of an overt deictic marker, Deix would simply be unspecified for author.

- (54) Mets les livres *là* a_i =dedans t_i .
'Put the books inside there.'

5 Conclusion

After establishing that implicit arguments in prepositional contexts are not a heterogeneous class, this chapter focused on one particular type of implicit argument: null pronouns (*pro*). Through the lens of a three-way typology (weak prepositions, strong prepositions, and axial part complexes), we demonstrated that this formal approach captures important generalizations about the licensing of null pronouns in SPEX.

Spanish and French converge in terms of weak prepositions; the inventories of weak prepositions are similar, and null pronouns are prohibited on semantic grounds. They also converge in licensing null pronouns with axial part complexes. Based on what appear to be much looser constraints on licensing null wholes in part-whole relations (see Section 4.1.2), this convergence is not surprising. The two languages converge again in both having a set of obligate orphan prepositions with deictic semantics but differing morphology (from an etymological perspective). The unique point of variation is in the licensing of null pronouns with strong prepositions. They are a locus of variation in at least two respects: First, there is variation within French; strong prepositions show mixed behaviour in that some of them allow and others prohibit null pronouns. Second, there is variation between French and Spanish since in Spanish the prohibition on orphaning with strong prepositions is virtually categoric. *Para* 'for' is currently our only candidate for a strong preposition allowing null pronouns in Spanish, and even *para* is a doubtful candidate since (i) native speakers' judgments on its use with a null pronoun vary and (ii) it is also the equivalent of French *pour* for which Authier (2016) argues its implicit argument is not syntactically projected (and hence not a null pronoun). The link between preposition type and the availability of null pronouns is schematically summarized in Tables 1 and 2.

²⁸ This analysis of *ci* and *là* can be naturally extended to their postnominal distribution (*ce livre-là* 'that book there') as adjunct PPs, diverging from Bernstein (1997) according to which post-nominal *là* is derived via N-raising, and Kayne & Pollock (2010) who treat post nominal *ci* and *là* as reduced relatives. Treating *ci* and *là* as instantiations of spatial deixis and person might also nicely capture previous treatments of the demonstratives *cela* and *ceci* (Kayne & Pollock 2010) and strong *là* as a DP (Authier 2016: 245-248).

Overt argument	ok				*
Null pronoun	*		ok		
Examples	<i>à, de, en</i>	<i>parmi, chez</i>	<i>avec, contre</i>	<i>en face (de), à côté (de)</i>	<i>(là-)dedans, (ci-)dessous</i>
Preposition type	weak	strong		axial part complex	

Table 1: Preposition types and the availability of null pronouns and overt arguments in French

Overt argument	ok			*
Null pronoun	*		ok	
Examples	<i>a, de, en</i>	<i>con, contra</i>	<i>encima (de), detrás (de)</i>	<i>abajo, atrás</i>
Preposition type	weak	strong	axial part complex	

Table 2: Preposition types and the availability of null pronouns and overt arguments in Spanish

This brief description of French and Spanish orphaning demonstrates, we believe, the enduring empirical and theoretical promise of comparative research in the prepositional domain, particularly through the lens of implicit arguments.

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